

NZ AI Policy Sandbox – Model Equations v0.3

All 19 ANZSIC Sectors: Calibrated Parameter Values

New Zealand Whole-Economy Policy Sandbox

AI for Good New Zealand

Version 0.3 – April 2026

Abstract

This document extends the NZ AI Policy Sandbox (NZAIS) model equations from v0.2 (structural specification, all 19 sectors) to v0.3 (calibrated specification). Every parameter introduced in v0.1 and v0.2 now carries a numerical value, an evidence class (observed, derived, or assumed), a primary source, and a confidence level (high, medium, or low).

v0.3 does not build or simulate the model. The output is a calibrated mathematical specification suitable for implementation. Equation forms are unchanged from v0.2. The new material in v0.3 consists entirely of Sections 9–15: parameter tables for GDP share weights, the economy-wide enabling stock, all nine Tier 1 sectors, all six Tier 2 sectors, all four Tier 3 sectors, and the confidence-adjusted aggregate.

Where evidence is weak, transparent assumptions are used with explicit sensitivity ranges rather than spurious point estimates. Every parameter without an observed NZ source is labelled assumed or derived and carries a confidence tag. No parameter is hidden or silent.

The underlying equation structure is inherited from v0.2 and is unchanged. Sections 1–8 of this document reproduce the v0.2 structural specification verbatim for self-contained reference.

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1 Model Structure

1.1 Design Principle: One Mechanism Per Layer

The model follows a strict causal chain where each variable is responsible for one layer. Each arrow in the DAG below represents exactly one mechanism and appears nowhere else in the system.

$$\begin{array}{ccccccc}
 E & \rightarrow & K_s & \rightarrow & A_s & \rightarrow & P_s \rightarrow L_s \\
 & & \uparrow & & & & \\
 & & \text{feedback from } A_s & & & &
 \end{array}$$

- E does not act directly on adoption – it acts *through* K_s (enabling capacity raises absorptive capability) for Tier 1, or through the fixed absorptive multiplier β_s for Tier 2 and Tier 3.
- K_s (Tier 1) or $\beta_s E$ (Tier 2/3) does not create productivity directly – it enables adoption *through* A_s .
- A_s does not create labour pressure directly in Tier 1 – it creates productivity *through* P_s , and labour pressure emerges *through* L_s .
- P_s is driven by adoption; L_s is driven by adoption and sector structure (not by P_s directly, to avoid double counting).
- There is one feedback in Tier 1: adoption A_s reinforces the demand for capability K_s (learning-by-doing, procurement maturity). This feedback does not appear in Tier 2 or Tier 3.

No mechanism appears twice. Every causal path is traceable through a single logical chain.

1.2 Tier Structure and State Reduction

The 19 ANZSIC Level 1 sectors are organised into three tiers, each with a different level of modelling fidelity. This tiering is a modelling decision, not an empirical finding. It reflects data availability, analytical priority, and the principle of tractability over false precision.

Tier	Sectors	State	Rationale
Tier 1	9 sectors	(K_s, A_s, P_s, L_s)	Full causal chain. Evidence base sufficient for sector archetypes. Covers approximately 61% of GDP.
Tier 2	6 sectors	(A_s, P_s)	Reduced fidelity. Absorptive capability replaced by fixed multiplier. Labour reported ex-post only.
Tier 3	4 sectors	A_s	Minimal state. Included for denominator completeness. Productivity and labour are ex-post only.

The tier reductions make three structural choices, each a modelling decision:

1. **Tier 2 capability reduction.** The dynamic K_s equation is replaced by a fixed sector absorptive multiplier $\beta_s \in [0, 1]$ applied to the shared enabling stock E . This collapses the two-layer $E \rightarrow K_s \rightarrow A_s$ path into a one-layer $E \rightarrow A_s$ path for Tier 2. The structural coupling to E is preserved; the dynamic accumulation is dropped.
2. **Tier 2 labour reporting.** L_s is not a dynamic state for Tier 2. Labour adjustment pressure is reported ex-post as $L_s^{T2} = \phi_s A_s$ where $\phi_s \in [0, 1]$ is a sector labour-exposure coefficient. It does not feed back into any dynamic equation.
3. **Tier 3 state minimisation.** Only adoption A_s is dynamic. Productivity is reported ex-post as $P_s^{T3} = \psi_s A_s$ where $\psi_s \in [0, 1]$. No labour reporting. Tier 3 sectors are included solely for denominator honesty in the whole-economy aggregate.

The AGENTS-level principle that sector blocks – not firm microstates – are the unit of analysis applies at all tiers. Each tier-block is diagonal: sectors interact only through the shared $E(t)$. No sector-pair spillovers are modelled at any tier.

1.3 State Variables

Definition 1 (Tier 1 Sector State Variables). *For each sector $s \in \mathcal{S}_1$ (Tier 1, $|\mathcal{S}_1| = 9$), the model tracks four state variables, all bounded in $[0, 1]$:*

$$K_s(t) : \text{Absorptive capability – effective capacity to absorb AI into productive use} \quad (1)$$

$$A_s(t) : \text{AI adoption maturity – degree of meaningful operational AI deployment} \quad (2)$$

$$P_s(t) : \text{Realised productivity effect – normalised realisation index, bounded in } [0, 1] \quad (3)$$

$$L_s(t) : \text{Labour adjustment pressure – displacement and reconfiguration index} \quad (4)$$

Definition 2 (Tier 2 Sector State Variables). *For each sector $s \in \mathcal{S}_2$ (Tier 2, $|\mathcal{S}_2| = 6$), the model tracks two dynamic state variables:*

$$A_s(t) : \text{AI adoption maturity – bounded in } [0, 1] \quad (5)$$

$$P_s(t) : \text{Realised productivity effect – bounded in } [0, 1] \quad (6)$$

Two additional quantities are reported ex-post but are not dynamic states:

$$L_s^{T2}(t) = \phi_s A_s(t) : \text{Labour adjustment pressure (reported only)} \quad (7)$$

$$\beta_s \in [0, 1] : \text{Fixed absorptive multiplier (parameter, not a state)} \quad (8)$$

Definition 3 (Tier 3 Sector State Variables). *For each sector $s \in \mathcal{S}_3$ (Tier 3, $|\mathcal{S}_3| = 4$), the model tracks one dynamic state variable:*

$$A_s(t) : \text{AI adoption maturity – bounded in } [0, 1] \quad (9)$$

One additional quantity is reported ex-post:

$$P_s^{T3}(t) = \psi_s A_s(t) : \text{Realised productivity effect (reported only)} \quad (10)$$

$\gamma_s \in [0, 1]$ is a fixed adoption multiplier (parameter, not a state). No labour reporting for Tier 3.

Definition 4 (Economy-Wide State Variable). *There is one economy-wide enabling state, shared across all tiers:*

$$E(t) : \text{National enabling capacity – economy-wide environment for AI adoption} \quad (11)$$

bounded in $[0, 1]$.

Remark 1. $P_s(t) \in [0, 1]$ by normalisation to the sector-specific productivity ceiling $\bar{p}_s$. The raw productivity gain is $\bar{p}_s \cdot P_s(t)$, where $\bar{p}_s$ is calibrated in Section 12.

Remark 2. The GDP share ω_s for each sector is a constant input parameter, not a state variable. It is sourced from `data/sector-parameter-table.csv` (Stats NZ GDP by Industry). $\sum_{s \in \mathcal{S}} \omega_s = 1$ across all 19 sectors. Numerical values are provided in Section 10.

Tier 1 Sectors

The nine Tier 1 sectors are:

Index	Sector	Symbol
1	Agriculture	$s = \text{ag}$
2	Manufacturing	$s = \text{mfg}$
3	Professional Services	$s = \text{prof}$
4	Public Sector	$s = \text{pub}$
5	Technology	$s = \text{tech}$
6	Healthcare	$s = \text{hlth}$
7	Construction	$s = \text{con}$
8	Financial Services	$s = \text{fin}$
9	Retail & Wholesale	$s = \text{ret}$

Tier 2 Sectors

The six Tier 2 sectors are:

Index	Sector	Symbol
10	Education & Training	$s = \text{edu}$
11	Transport & Warehousing	$s = \text{trn}$
12	Accommodation & Food Services	$s = \text{acc}$
13	Administrative & Support Services	$s = \text{adm}$
14	Information Media & Telecommunications	$s = \text{imt}$
15	Utilities	$s = \text{utl}$

Tier 3 Sectors

The four Tier 3 sectors are:

Index	Sector	Symbol
16	Mining	$s = \text{min}$
17	Rental, Hiring & Real Estate	$s = \text{rnt}$
18	Arts & Recreation	$s = \text{art}$
19	Other Services	$s = \text{oth}$

2 Tier 1 Core Dynamics

Tier 1 equations are identical to v0.1. Do not modify.

Assumption 1 (Nonnegative Rates). *All rate parameters are nonnegative:*

$$\rho_K, \rho_A, \rho_P, \rho_L, \rho_E, \alpha_s, \kappa_s, \phi_s, \lambda_s, \eta_s \geq 0.$$

This ensures the bounded gain-loss forms keep all state variables in $[0, 1]$ by construction, without requiring ex-post clamping.

All equations use bounded gain-loss forms:

$$\frac{dX}{dt} = (1 - X) \cdot \rho_X \cdot f(\text{drivers}) - X \cdot \rho_X \cdot g(\text{drivers})$$

so $X \in [0, 1]$ is maintained for all $t \geq 0$. See Section 7 for the formal proposition.

2.1 Capability Stock Dynamics – $K_s(t)$

Absorptive capability grows when the national enabling environment is strong and when sector-specific support is provided. It decays in the absence of investment and maintenance.

$$\boxed{\frac{dK_s}{dt} = (1 - K_s) \cdot \rho_K \cdot (\phi_s E + G_s + \eta_s A_s) - K_s \cdot \rho_K \cdot \mu_s} \quad (12)$$

Interpretation:

- $(1 - K_s) \cdot \rho_K \cdot \phi_s E$: national enabling capacity lifts sector capability, at rate proportional to the sector's exposure ϕ_s to the national environment.
- $(1 - K_s) \cdot \rho_K \cdot G_s$: sector-specific policy support G_s directly builds capability.
- $(1 - K_s) \cdot \rho_K \cdot \eta_s A_s$: adoption-driven demand reinforces capability building (the one feedback in the DAG). $\eta_s \geq 0$ is the adoption feedback parameter.
- $K_s \cdot \rho_K \cdot \mu_s$: capability decays at rate μ_s (skill attrition, technology obsolescence, institutional drift).
- $(1 - K_s)$ saturation prevents $K_s > 1$; K_s decay prevents $K_s < 0$.

Remark 3. G_s is a policy control input (exogenous), not a state variable. In the base scenario (no intervention), $G_s = 0$.

2.2 AI Adoption Dynamics – $A_s(t)$

Adoption grows when capability is high and decays in the absence of enabling conditions.

$$\boxed{\frac{dA_s}{dt} = (1 - A_s) \cdot \rho_A \cdot \alpha_s K_s - A_s \cdot \rho_A \cdot (1 - K_s)} \quad (13)$$

Interpretation:

- $(1 - A_s) \cdot \rho_A \cdot \alpha_s K_s$: adoption expands when capability is available. $\alpha_s \geq 0$ is the adoption responsiveness to capability for sector s .
- $A_s \cdot \rho_A \cdot (1 - K_s)$: when capability is weak, existing adoption stalls or reverses.

Remark 4. E does not appear directly in dA_s/dt . The national enabling environment acts only through K_s . This preserves the one-mechanism-per-layer principle.

2.3 Productivity Dynamics – $P_s(t)$

Realised productivity grows as adoption matures. It decays slowly when adoption is low.

$$\boxed{\frac{dP_s}{dt} = (1 - P_s) \cdot \rho_P \cdot \kappa_s A_s - P_s \cdot \rho_P \cdot (1 - A_s)} \quad (14)$$

Interpretation:

- $(1 - P_s) \cdot \rho_P \cdot \kappa_s A_s$: realised productivity gains accumulate with adoption. $\kappa_s \geq 0$ is the productivity responsiveness to adoption for sector s .
- $P_s \cdot \rho_P \cdot (1 - A_s)$: when adoption falls, the realised gain decays back toward zero.

Remark 5. K_s does not appear directly in dP_s/dt . Capability affects productivity only through A_s . This is the one-mechanism-per-layer constraint.

2.4 Labour Adjustment Pressure – $L_s(t)$

Labour adjustment pressure rises as adoption expands, modulated by sector-specific labour sensitivity.

$$\boxed{\frac{dL_s}{dt} = (1 - L_s) \cdot \rho_L \cdot \lambda_s A_s - L_s \cdot \rho_L \cdot (1 - A_s)} \quad (15)$$

Interpretation:

- $(1 - L_s) \cdot \rho_L \cdot \lambda_s A_s$: pressure rises with adoption. $\lambda_s \geq 0$ captures sector-specific labour sensitivity.
- $L_s \cdot \rho_L \cdot (1 - A_s)$: pressure eases as adoption plateaus or reverses.

Remark 6. P_s does not appear in dL_s/dt . Labour pressure is driven by adoption, not directly by productivity gains. This avoids double counting.

2.5 National Enabling Capacity – $E(t)$

$E(t)$ is the single economy-wide state. It grows with national investment G_E and decays with depreciation.

$$\boxed{\frac{dE}{dt} = (1 - E) \cdot \rho_E \cdot G_E - E \cdot \rho_E \cdot \delta_E} \quad (16)$$

Interpretation:

- $(1 - E) \cdot \rho_E \cdot G_E$: national enabling investment G_E lifts the economy-wide enabling stock.
- $E \cdot \rho_E \cdot \delta_E$: the enabling environment depreciates at rate δ_E .

Remark 7. E acts on the whole economy simultaneously – on all 19 sectors across all three tiers. In the base scenario (no supply-side intervention), $G_E = G_E^{base} \geq 0$.

3 Tier 2 Dynamics

Tier 2 sectors use a reduced state (A_s, P_s) . The dynamic K_s equation from Tier 1 is replaced by a fixed sector absorptive multiplier $\beta_s \in [0, 1]$ applied to the economy-wide enabling stock. L_s is reported ex-post as a linear function of adoption; it is not a dynamic state.

Assumption 2 (Tier 2 Nonnegative Parameters). All Tier 2 rate parameters are nonnegative:

$$\rho_A, \rho_P \geq 0, \quad \beta_s, \kappa_s, \phi_s \in [0, 1].$$

3.1 Tier 2 Causal Structure

The Tier 2 DAG collapses the $E \rightarrow K_s \rightarrow A_s$ two-step into a single $E \rightarrow A_s$ step:

$$E \rightarrow A_s \rightarrow P_s$$

The absorptive multiplier β_s captures the sector's structural coupling to the national enabling environment. It is a fixed parameter, not a dynamic state, and does not accumulate or decay over time.

3.2 Tier 2 AI Adoption Dynamics – $A_s(t)$

For Tier 2 sectors, the enabling environment E acts directly on adoption, scaled by the sector absorptive multiplier β_s :

$$\boxed{\frac{dA_s}{dt} = (1 - A_s) \cdot \rho_A \cdot \beta_s E - A_s \cdot \rho_A \cdot (1 - \beta_s E)} \quad (17)$$

Remark 8. Since $\beta_s \in [0, 1]$ and $E \in [0, 1]$, the product $\beta_s E \in [0, 1]$ by construction. This ensures the loss driver $(1 - \beta_s E) \in [0, 1]$, which is required for the bounded gain-loss form to maintain $A_s \in [0, 1]$.

3.3 Tier 2 Productivity Dynamics – $P_s(t)$

Tier 2 productivity follows the same form as Tier 1, driven by adoption:

$$\boxed{\frac{dP_s}{dt} = (1 - P_s) \cdot \rho_P \cdot \kappa_s A_s - P_s \cdot \rho_P \cdot (1 - A_s)} \quad (18)$$

3.4 Tier 2 Labour Reporting (Ex-Post)

Labour adjustment pressure for Tier 2 sectors is *not* a dynamic state. It is reported ex-post as a linear function of adoption:

$$L_s^{T2}(t) = \phi_s \cdot A_s(t), \quad s \in \mathcal{S}_2 \quad (19)$$

where $\phi_s \in [0, 1]$ is the sector labour-exposure coefficient.

Remark 9. In Tier 1, ϕ_s denotes the enabling exposure of sector s to the national environment (coefficient on E in Equation (12)). In Tier 2, ϕ_s denotes the sector labour-exposure coefficient in the ex-post expression (19). Because Tier 1 and Tier 2 sectors are disjoint, the symbol ϕ_s is unambiguous for any given sector s .

4 Tier 3 Dynamics

Tier 3 sectors use a minimal state: adoption A_s only. These sectors are included for denominator honesty in the whole-economy aggregate.

Assumption 3 (Tier 3 Nonnegative Parameters). *All Tier 3 parameters are nonnegative:*

$$\rho_A \geq 0, \quad \gamma_s, \psi_s \in [0, 1].$$

4.1 Tier 3 AI Adoption Dynamics – $A_s(t)$

$$\boxed{\frac{dA_s}{dt} = (1 - A_s) \cdot \rho_A \cdot \gamma_s E - A_s \cdot \rho_A \cdot (1 - \gamma_s E)} \quad (20)$$

4.2 Tier 3 Productivity Reporting (Ex-Post)

Productivity for Tier 3 sectors is *not* dynamic. It is reported ex-post as:

$$P_s^{T3}(t) = \psi_s \cdot A_s(t), \quad s \in \mathcal{S}_3 \quad (21)$$

where $\psi_s \in [0, 1]$ is the sector productivity coefficient.

No labour reporting for Tier 3. These sectors are included in the model for denominator honesty only.

5 Policy Scenarios

All three scenarios share the same equal budget $\Delta \geq 0$:

$$\Delta_A = \Delta_B = \Delta_C = \Delta.$$

This is required for like-for-like comparison. Let $n = |\mathcal{S}| = 19$ (all sectors across all tiers).

5.1 Scenario A – Aggregate Policy

Policy support is distributed uniformly across all 19 sectors by equal per-sector allocation.

$$\begin{aligned} \text{Tier 1: } \frac{dK_s}{dt} &= (1 - K_s) \cdot \rho_K \cdot \left(\phi_s E + \frac{\Delta_A}{n} + \eta_s A_s \right) - K_s \cdot \rho_K \cdot \mu_s \\ \text{Tier 2: } \frac{dA_s}{dt} &= (1 - A_s) \cdot \rho_A \cdot \left(\beta_s E + \frac{\Delta_A}{n} \right) - A_s \cdot \rho_A \cdot (1 - \beta_s E) \\ \text{Tier 3: } \frac{dA_s}{dt} &= (1 - A_s) \cdot \rho_A \cdot \left(\gamma_s E + \frac{\Delta_A}{n} \right) - A_s \cdot \rho_A \cdot (1 - \gamma_s E) \end{aligned} \quad (22)$$

5.2 Scenario B – Targeted Demand-Side Policy

Policy support is weighted toward sectors with greater adoption gaps.

$$w_s^{\text{demand}} = \frac{1 - A_s(0)}{\sum_{s' \in \mathcal{S}} (1 - A_{s'}(0))}, \quad \forall s \in \mathcal{S} \quad (23)$$

so that $\sum_{s \in \mathcal{S}} w_s^{\text{demand}} = 1$ and the total budget is exactly Δ_B .

5.3 Scenario C – Targeted Supply-Side Policy

Policy support is concentrated entirely on the national enabling stock:

$$G_E^{(C)} = G_E^{\text{base}} + \Delta_C, \quad G_s^{(C)} = 0 \quad \forall s \in \mathcal{S}. \quad (24)$$

6 Aggregation

6.1 Whole-Economy Productivity Aggregate

$$\bar{P}(t) = \sum_{s \in \mathcal{S}_1} \omega_s P_s(t) + \sum_{s \in \mathcal{S}_2} \omega_s P_s(t) + \sum_{s \in \mathcal{S}_3} \omega_s \psi_s A_s(t) \quad (25)$$

where ω_s is the sector's share of whole-economy GDP, with $\sum_{s \in \mathcal{S}} \omega_s = 1$.

7 Boundedness

Proposition 1 (State Variables Remain Bounded Across All Tiers). *Suppose all parameters satisfy Assumptions 1, 2, and 3 (nonnegative rates; $\beta_s, \gamma_s, \psi_s, \phi_s \in [0, 1]$). If all dynamic state variables satisfy their initial conditions in $[0, 1]$:*

$$K_s(0), A_s(0), P_s(0), L_s(0), E(0) \in [0, 1] \quad \forall s \in \mathcal{S}_1,$$

$$A_s(0), P_s(0) \in [0, 1] \quad \forall s \in \mathcal{S}_2,$$

$$A_s(0) \in [0, 1] \quad \forall s \in \mathcal{S}_3,$$

then for all $t \geq 0$ all state variables remain in $[0, 1]$. Furthermore, the ex-post quantities $L_s^{T2}(t) = \phi_s A_s(t)$ and $P_s^{T3}(t) = \psi_s A_s(t)$ are also in $[0, 1]$ for all $t \geq 0$.

Proof. Each equation has the bounded gain-loss form $\dot{X} = (1 - X)f^+(X) - Xf^-(X)$ where $f^+, f^- \geq 0$ by the nonnegative parameter assumptions. At $X = 1$: $\dot{X} = -f^-(1) \leq 0$. At $X = 0$: $\dot{X} = f^+(0) \geq 0$. Hence $[0, 1]$ is positively invariant for each equation. The ex-post quantities are products of two $[0, 1]$ -valued quantities and are therefore also in $[0, 1]$. The full argument is given in the v0.2 proof. $\square$

8 Known Limits Inherited from v0.2

The following structural limits from v0.2 carry over unchanged to v0.3.

1. **Tier 2 and Tier 3 state reductions are modelling choices, not empirical findings.**
2. **No sector-pair spillovers.** Sectors interact only through the shared enabling stock $E(t)$.
3. **Deterministic ODE only.** No stochastic shocks, no uncertainty propagation, no Monte Carlo.
4. **Single-country model.** No trade terms, international spillovers, or cross-border adoption dynamics.
5. **Tier 2 and Tier 3 labour effects are reported-only, not dynamic.**
6. **Policy controls are constant within a scenario.** G_s and G_E are treated as fixed scenario parameters, not adaptive policies.

9 v0.3 Calibration: Approach and Evidence Classes

Every numerical value introduced in v0.3 carries three mandatory tags:

Tag	Meaning
Evidence class	observed: directly supported by NZ data. derived: estimated from multiple sources or international benchmarks. assumed: transparent placeholder, no direct evidence base.
Source	Primary source citation for the value or the reasoning behind it.
Confidence	high: narrow uncertainty range, strong NZ evidence. medium: reasonable but imprecise NZ evidence or good international evidence. low: sparse evidence, wide sensitivity range.

Calibration philosophy. Where NZ-specific evidence exists (observed or derived), it is used. Where it does not, international benchmarks from the OECD, AI Forum NZ, Datacom, and sector-specific sources are used to derive plausible ranges. Parameters that remain unsupported are labelled assumed and carry a stated sensitivity range. No parameter is spuriously precise.

No time-series fitting. v0.3 calibration uses cross-sectional evidence per sector. It does not fit parameters to historical time series. That is v0.4 or later work.

Shared global rates. The rate parameters $\rho_K, \rho_A, \rho_P, \rho_L$ are shared across sectors within each tier. Sector heterogeneity is captured entirely by the sector-specific structural parameters ($\alpha_s, \kappa_s, \lambda_s, \phi_s, \eta_s, \mu_s$ for Tier 1; $\beta_s, \kappa_s, \phi_s$ for Tier 2; γ_s, ψ_s for Tier 3). This maintains model parsimony.

10 v0.3 Calibration: GDP Share Weights

GDP shares ω_s are the primary weighting mechanism for the whole-economy aggregate $\bar{P}(t)$. They must sum to exactly 1.

10.1 Sector GDP Mapping

The model's 19 sectors do not correspond one-to-one to the 19 ANZSIC Level 1 categories. Three mapping adjustments are required:

1. **S03 Professional Services** in the parameter table uses a combined GDP figure that includes the ANZSIC Administrative and Support Services category. For non-overlapping allocation, the combined value (\$32,757M) is split into S03 (Professional, Scientific and Technical Services portion: \$21,000M) and S13 (Administrative and Support Services portion: \$11,757M) based on ANZSIC relative shares.
2. **S05 Technology** is a cross-sectoral category (NZTech definition: ICT + hi-tech manufacturing + biotech). The allocated GDP (\$12,434M) is derived as NZTech sector total (\$24,000M) minus the ANZSIC Information Media and Telecommunications sector (\$11,566M, captured in S14). This removes the primary telecomm overlap.
3. **S18 Arts and Recreation** and **S19 Other Services** are split equally from the combined parameter table value (\$8,778M), each receiving \$4,389M, consistent with ANZSIC relative proportions.

All other sectors use the GDP values directly from `data/sector-parameter-table.csv` (Stats NZ GDP by Industry, chain-volume, YE March 2025).

10.2 GDP Share Table

The model-consistent total GDP used as the denominator is \$281,020M. This is the sum of the 19 allocated sector GDPs defined above.

Sector	GDP (NZD M)	ω_s	Class	Source	Conf.
S01 Agriculture	14,396	0.0512	observed	Stats NZ GDP by Industry	high
S02 Manufacturing	22,147	0.0788	observed	Stats NZ GDP by Industry	high
S03 Professional Services	21,000	0.0747	derived	Stats NZ ANZSIC split	medium
S04 Public Sector	13,499	0.0480	observed	Stats NZ GDP by Industry	high
S05 Technology	12,434	0.0442	derived	NZTech minus S14 overlap	medium
S06 Healthcare	19,175	0.0682	observed	Stats NZ GDP by Industry	high
S07 Construction	18,085	0.0643	observed	Stats NZ GDP by Industry	high
S08 Financial Services	15,937	0.0567	observed	Stats NZ GDP by Industry	high
S09 Retail & Wholesale	32,376	0.1152	observed	Stats NZ GDP by Industry	high
S10 Education	10,127	0.0360	observed	Stats NZ GDP by Industry	high
S11 Transport/Warehousing	11,651	0.0415	observed	Stats NZ GDP by Industry	high
S12 Accommodation/Food	8,720	0.0310	derived	Stats NZ ANZSIC est.	medium
S13 Admin/Support Services	11,757	0.0418	derived	Stats NZ ANZSIC split	medium
S14 Info Media/Telecomms	11,566	0.0412	observed	Stats NZ GDP by Industry	high
S15 Utilities	7,032	0.0250	observed	Stats NZ GDP by Industry	high
S16 Mining	2,024	0.0072	observed	Stats NZ GDP by Industry	high
S17 Rental/Real Estate	40,316	0.1435	observed	Stats NZ GDP by Industry	high
S18 Arts & Recreation	4,389	0.0156	derived	Stats NZ ANZSIC split	medium

Sector	GDP (NZD M)	ω_s	Class	Source	Conf.
S19 Other Services	4,389	0.0156	derived	Stats NZ ANZSIC split	medium
Total	281,020	1.0000			

Remark 10. Tier 1 sectors (S01–S09) account for $\sum_{s \in \mathcal{S}_1} \omega_s = 0.506$ (50.6% of the model-consistent GDP). Tier 2 (S10–S15) accounts for 0.217 (21.7%), and Tier 3 (S16–S19) accounts for 0.282 (28.2%). The Tier 3 weight is elevated by the large Rental and Real Estate sector (S17, $\omega = 0.1435$), which reflects the dominance of imputed owner-occupied rent in NZ national accounts.

11 v0.3 Calibration: Economy-Wide Enabling Stock

The enabling stock $E(t)$ is the single economy-wide state variable shared across all 19 sectors. Its initial value and dynamics reflect NZ’s current enabling environment for AI adoption: infrastructure, skills, policy, digital connectivity.

Parameter	Value	Unit	Class	Source	Conf.
$E(0)$	0.38	$[0, 1]$	assumed	NZ mid-tier OECD positioning [2, 3]	low
ρ_E	0.25	yr^{-1}	assumed	Consistent with 4-year adjustment horizon	low
δ_E	0.05	yr^{-1}	assumed	Enabling stock depreciates slowly	low
G_E^{base}	0.05	–	assumed	Baseline national investment rate	low

Calibration note. $E(0) = 0.38$ reflects NZ’s position as a mid-tier digital economy. The OECD analysis identifies NZ as lagging behind advanced economies in technology diffusion speed, intangible capital investment, and AI talent supply [2]. International comparators with strong enabling environments (US, Singapore, South Korea) would be estimated at $E(0) \approx 0.55$ –0.65. NZ’s known structural weaknesses – R&D at 0.8% of GDP (OECD average 1.8%), slow technology diffusion, ICT skills shortage – justify the below-OECD-average initial value.

Sensitivity range: $E(0) \in [0.28, 0.48]$.

At steady state with no intervention ($G_s = 0$, $G_E = G_E^{\text{base}}$), the long-run enabling stock converges to:

$$E^* = \frac{G_E^{\text{base}}}{G_E^{\text{base}} + \delta_E} = \frac{0.05}{0.10} = 0.50.$$

This places the NZ enabling environment at approximately 50% of its theoretical maximum under current policy, consistent with the medium-term potential identified in the NZ Treasury analysis.

12 v0.3 Calibration: Tier 1 Parameters

12.1 Tier 1 Global Rates

The following rates apply to all nine Tier 1 sectors. Sector heterogeneity is captured by the sector-specific structural parameters.

Parameter	Value	Unit	Class	Conf.
ρ_K	0.30	yr ⁻¹	assumed	low
ρ_A	0.35	yr ⁻¹	assumed	low
ρ_P	0.25	yr ⁻¹	assumed	low
ρ_L	0.35	yr ⁻¹	assumed	low

Rate calibration note. The rate $\rho_X = 0.30$ implies a time constant of approximately 3.3 years for a half-saturation response when the driving term is held constant. This is consistent with the Treasury finding that AI diffusion in advanced economies typically takes 3–10 years for material productivity effects to emerge [3]. The 20–30 year full diffusion timeline noted in the Treasury analysis is captured by the system reaching higher adoption states over multiple time periods, not by setting lower rates.

12.2 Tier 1 Sector Calibration Notes

Agriculture ($s = \text{ag}$). The most evidentially constrained Tier 1 sector. No published NZ AI adoption rate exists. Halter (precision dairy AI, \$1.55B valuation, 1,000+ farmers) is the leading commercial deployment but represents a small fraction of the sector’s 61,593 enterprises [10]. Long diffusion timelines (5–10 years to scale) and low baseline digital maturity justify the lowest initial conditions of any Tier 1 sector. The OECD breakthrough pathway – AI-robotics integration in primary industries – is represented by the sector’s upside sensitivity range.

Manufacturing ($s = \text{mfg}$). 58% adoption in large firms (Datacom 2024, 100+ employee firms) [5], but economy-wide adoption including SMEs is substantially lower. Callaghan Innovation’s Industry 4.0 framework and the Investment Boost (20% tax deduction) are active enablers. Productivity potential of 15–30% per the OECD analysis [2]. Second-highest α_s after Technology.

Professional Services ($s = \text{prof}$). Highest current adoption of any sector. Thomson Reuters 2025 reports GenAI active use doubled from 12% to 22% in 12 months. RBNZ identifies professional and managerial occupations as the highest AI exposure group [7]. The sector faces diminishing returns from further investment given the already-high adoption base: κ_{prof} is set lower than manufacturing despite the higher $A_s(0)$.

Public Sector ($s = \text{pub}$). Lowest adoption base but fastest acceleration trajectory. GCDO AI Use Case Census 2025 records 272 use cases across 70 agencies, with 55 fully operational (up from 15 the previous year) [4]. Capability gap and risk-averse institutional culture (the “chilling effect” from early guidance) are the primary barriers. The Public Service AI Framework (January 2025) provides structural support for accelerating adoption.

Technology ($s = \text{tech}$). Supply-side enabler sector. Very high initial state ($A_{\text{tech}}(0) = 0.65$, $K_{\text{tech}}(0) = 0.72$) reflects that the Technology sector is both a major AI user and the primary producer of AI capabilities. The highest $\phi_{\text{tech}} = 0.75$ reflects the Technology sector’s exceptional structural coupling to the national enabling environment. TIN200 revenue growing 9.9% indicates continued expansion despite first workforce contraction since the GFC [9].

Healthcare ($s = \text{hlth}$). 62% adoption in large firms (Datacom 2024) [5] blurs back-office AI and clinical AI. The PMCSA analysis distinguishes between admin AI (low-hanging fruit, faster realisation) and clinical AI (slow deployment due to equity, governance, and interoperability constraints) [6]. The 85% of health systems that do not support data sharing is a binding structural constraint on $K_{\text{hlth}}(0)$.

Construction ($s = \text{con}$). The evidentially weakest Tier 1 sector. No published NZ AI adoption rate. 95% of firms have fewer than 10 employees, and digital maturity is the lowest of any major sector. However, AI is entering the sector’s top-5 concerns for the first time (BDO 2025) and the \$120B government infrastructure pipeline creates demand-side pull. NZ case study: Classic Group (Tauranga) reduced purchase order pricing from 2 hours to 15 seconds using AWS Bedrock + Claude Sonnet.

Financial Services ($s = \text{fin}$). High adoption constrained by regulation rather than capability. The FMA survey (9/13 respondents using AI) [8] has an inadequate sample for sector-level inference; confidence is low. Banking is outpacing insurance. RBNZ identifies a regulatory vacuum from 2025–2028 as the primary short-term risk [7]. Productivity potential 15–25% cost reduction per the OECD.

Retail and Wholesale ($s = \text{ret}$). Combined sector. Wholesale scored the highest AI adoption rate of any NZ sector in Datacom 2024 (64%) [5]; retail is substantially lower and more politically sensitive (displacement story). The combined initial condition $A_{\text{ret}}(0) = 0.35$ is a weighted average. Labour sensitivity λ_{ret} is set high (0.55) given the large low-wage workforce with automation exposure.

12.3 Tier 1 Per-Sector Parameter Tables

For each sector, the table format is:

- Initial conditions: $K_s(0)$, $A_s(0)$, $P_s(0)$, $L_s(0)$
- Structural parameters: α_s , κ_s , λ_s , ϕ_s , η_s , μ_s , $\bar{p}_s$
- Evidence class: obs = observed, der = derived, asm = assumed
- Confidence: H = high, M = medium, L = low

S01 – Agriculture

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{ag}}(0)$	0.12	[0.07, 0.18]	asm	Digital maturity proxy	L
$A_{\text{ag}}(0)$	0.05	[0.03, 0.10]	asm	No NZ adoption data; Halter proxy	L
$P_{\text{ag}}(0)$	0.03	[0.01, 0.06]	asm	Derived from $A_{\text{ag}}(0)$	L
$L_{\text{ag}}(0)$	0.04	[0.02, 0.07]	asm	Low automation exposure	L
α_{ag}	0.50	[0.35, 0.65]	der	Sector archetype: medium responsiveness	M
κ_{ag}	0.15	[0.10, 0.25]	der	OECD 10–25% productivity range	M
λ_{ag}	0.30	[0.20, 0.42]	asm	Physical work limits automation exposure	L
ϕ_{ag}	0.20	[0.12, 0.30]	asm	Limited rural enabling exposure	L
η_{ag}	0.25	[0.15, 0.38]	asm	Learning-by-doing (Halter model)	L
μ_{ag}	0.10	[0.06, 0.15]	asm	Moderate skill attrition	L
$\bar{p}_{\text{ag}}$	0.20	[0.10, 0.25]	der	OECD robotics-integration upside	M

S02 – Manufacturing

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{mfg}}(0)$	0.42	[0.32, 0.52]	asm	Moderate I4.0 digital maturity	M
$A_{\text{mfg}}(0)$	0.35	[0.25, 0.48]	der	58% large firms (Datacom); est. 35% economy-wide	M
$P_{\text{mfg}}(0)$	0.22	[0.15, 0.30]	asm	Derived from $A_{\text{mfg}}(0)$	M
$L_{\text{mfg}}(0)$	0.18	[0.12, 0.25]	asm	Moderate automation exposure	M
α_{mfg}	0.70	[0.55, 0.82]	der	I4.0 pathway high responsiveness	M
κ_{mfg}	0.22	[0.15, 0.30]	der	OECD 15–30% productivity range	M
λ_{mfg}	0.45	[0.32, 0.58]	asm	Significant machine-task automation	M
ϕ_{mfg}	0.40	[0.28, 0.52]	asm	Moderate enabling exposure	L
η_{mfg}	0.35	[0.22, 0.48]	asm	Learning-by-doing in I4.0 deployment	L
μ_{mfg}	0.08	[0.04, 0.12]	asm	Moderate skill depreciation	L
$\bar{p}_{\text{mfg}}$	0.25	[0.15, 0.30]	der	OECD manufacturing productivity range	M

S03 – Professional Services

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{prof}}(0)$	0.58	[0.48, 0.68]	asm	High digital maturity	M
$A_{\text{prof}}(0)$	0.45	[0.35, 0.55]	der	22% active GenAI (Thomson Reuters 2025); est. higher overall	M
$P_{\text{prof}}(0)$	0.28	[0.20, 0.36]	asm	Derived from $A_{\text{prof}}(0)$	M
$L_{\text{prof}}(0)$	0.22	[0.15, 0.30]	asm	High cognitive task exposure (RBNZ)	M
α_{prof}	0.80	[0.65, 0.92]	der	Knowledge work amplification	M
κ_{prof}	0.18	[0.12, 0.25]	der	Diminishing returns from high base	M
λ_{prof}	0.55	[0.40, 0.68]	asm	High cognitive task exposure (RBNZ)	M
ϕ_{prof}	0.55	[0.42, 0.68]	asm	High reliance on digital infrastructure	L
η_{prof}	0.45	[0.32, 0.58]	asm	Strong learning-by-doing feedback	L
μ_{prof}	0.07	[0.04, 0.11]	asm	Low skill attrition in knowledge sector	L
$\bar{p}_{\text{prof}}$	0.18	[0.10, 0.25]	der	OECD moderate; diminishing returns from high base	M

S04 – Public Sector

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{pub}}(0)$	0.15	[0.10, 0.22]	asm	GCDO capability gap evidence	M
$A_{\text{pub}}(0)$	0.10	[0.06, 0.15]	der	GCDO: 55 operational use cases / 272 total	M
$P_{\text{pub}}(0)$	0.06	[0.03, 0.09]	asm	Derived from $A_{\text{pub}}(0)$	M
$L_{\text{pub}}(0)$	0.08	[0.05, 0.12]	asm	Moderate public service AI exposure	L
α_{pub}	0.60	[0.45, 0.75]	der	Adoption follows capability with governance lag	M
κ_{pub}	0.12	[0.05, 0.15]	der	OECD 5–15% for public sector	M
λ_{pub}	0.40	[0.28, 0.52]	asm	Moderate public service exposure	L
ϕ_{pub}	0.35	[0.22, 0.48]	asm	Moderate enabling exposure, GCDO-dependent	L
η_{pub}	0.30	[0.18, 0.42]	asm	Some learning-by-doing in agency adoption	L
μ_{pub}	0.08	[0.04, 0.12]	asm	Moderate institutional skill drift	L
$\bar{p}_{\text{pub}}$	0.12	[0.05, 0.15]	der	OECD public sector range	M

S05 – Technology

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{tech}}(0)$	0.72	[0.60, 0.82]	asm	Highest digital maturity; NZTech Key Metrics	M
$A_{\text{tech}}(0)$	0.65	[0.55, 0.78]	der	Supply-side sector: very high internal AI adoption	M
$P_{\text{tech}}(0)$	0.40	[0.30, 0.50]	asm	Derived from $A_{\text{tech}}(0)$	M
$L_{\text{tech}}(0)$	0.30	[0.20, 0.42]	asm	High AI exposure in tech occupations	M
α_{tech}	0.90	[0.75, 1.00]	der	Technology sector: maximum responsiveness	M
κ_{tech}	0.25	[0.18, 0.35]	der	Strong internal productivity gains from AI tools	M
λ_{tech}	0.60	[0.45, 0.75]	asm	High cognitive/software task automation exposure	M
ϕ_{tech}	0.75	[0.60, 0.90]	asm	Technology IS the enabling sector	M
η_{tech}	0.60	[0.45, 0.75]	asm	Strongest learning-by-doing effect	L
μ_{tech}	0.05	[0.02, 0.09]	asm	Continuous learning maintains capability	L
$\bar{p}_{\text{tech}}$	0.30	[0.20, 0.40]	der	TIN200 growth rates proxy	M

S06 – Healthcare

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{hlth}}(0)$	0.38	[0.28, 0.48]	asm	Moderate; 85% interop barrier (PMCSA)	M
$A_{\text{hlth}}(0)$	0.35	[0.25, 0.48]	der	62% large firms (Datacom); blurs admin/clinical AI	M
$P_{\text{hlth}}(0)$	0.20	[0.14, 0.28]	asm	Derived from $A_{\text{hlth}}(0)$	M
$L_{\text{hlth}}(0)$	0.16	[0.10, 0.22]	asm	Augmentation-dominant sector	L
α_{hlth}	0.55	[0.40, 0.70]	der	Moderate, constrained by equity/governance	M
κ_{hlth}	0.15	[0.10, 0.20]	der	OECD 10–20%, admin AI higher than clinical	M
λ_{hlth}	0.35	[0.22, 0.48]	asm	Augmentation more than displacement	L
ϕ_{hlth}	0.35	[0.22, 0.48]	asm	Moderate enabling exposure, governance constraints	L
η_{hlth}	0.25	[0.15, 0.38]	asm	Some admin AI learning-by-doing	L
μ_{hlth}	0.08	[0.04, 0.12]	asm	Moderate skill attrition in health sector	L
$\bar{p}_{\text{hlth}}$	0.15	[0.10, 0.20]	der	OECD constrained by governance	M

S07 – Construction

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{con}}(0)$	0.08	[0.04, 0.14]	asm	Lowest digital maturity of any major sector	L
$A_{\text{con}}(0)$	0.06	[0.03, 0.12]	asm	No NZ adoption data; lowest maturity proxy	L
$P_{\text{con}}(0)$	0.03	[0.01, 0.06]	asm	Derived from $A_{\text{con}}(0)$	L
$L_{\text{con}}(0)$	0.05	[0.02, 0.09]	asm	Physical work: low near-term automation exposure	L
α_{con}	0.40	[0.25, 0.55]	der	Physical tasks limit AI application	L
κ_{con}	0.10	[0.05, 0.18]	der	Low near-term potential; digital maturity constraint	L
λ_{con}	0.25	[0.15, 0.38]	asm	Lower automation exposure in physical construction	L
ϕ_{con}	0.20	[0.10, 0.32]	asm	Limited enabling exposure; rural/dispersed sector	L
η_{con}	0.20	[0.10, 0.32]	asm	Limited learning-by-doing from low base	L
μ_{con}	0.10	[0.06, 0.15]	asm	High skill turnover; fragmented sector	L
$\bar{p}_{\text{con}}$	0.10	[0.05, 0.18]	der	Low near-term potential per OECD	L

S08 – Financial Services

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{fin}}(0)$	0.55	[0.42, 0.68]	asm	High digital infra; regulatory constraints	M
$A_{\text{fin}}(0)$	0.48	[0.32, 0.62]	der	FMA: 9/13 respondents; small sample (low conf.)	L
$P_{\text{fin}}(0)$	0.29	[0.18, 0.40]	asm	Derived from $A_{\text{fin}}(0)$	L
$L_{\text{fin}}(0)$	0.25	[0.16, 0.35]	asm	Moderate: regulatory constraints slow displacement	L
α_{fin}	0.75	[0.58, 0.88]	der	High when regulatory constraints permit	M
κ_{fin}	0.20	[0.15, 0.25]	der	OECD 15–25% cost reduction potential	M
λ_{fin}	0.50	[0.35, 0.65]	asm	High cognitive task exposure	M
ϕ_{fin}	0.50	[0.35, 0.65]	asm	High enabling exposure; digital-native sector	L
η_{fin}	0.40	[0.28, 0.52]	asm	Learning-by-doing in analytics	L
μ_{fin}	0.07	[0.03, 0.11]	asm	Low skill attrition in financial sector	L
$\bar{p}_{\text{fin}}$	0.22	[0.15, 0.25]	der	OECD cost reduction range	M

S09 – Retail and Wholesale

Parameter	Value	Sensitivity	Class	Source	Conf.
$K_{\text{ret}}(0)$	0.38	[0.28, 0.48]	asm	Wholesale more mature than retail	M
$A_{\text{ret}}(0)$	0.35	[0.25, 0.48]	der	Wholesale 64% (Datacom); retail lower; combined est.	M
$P_{\text{ret}}(0)$	0.20	[0.14, 0.28]	asm	Derived from $A_{\text{ret}}(0)$	M
$L_{\text{ret}}(0)$	0.20	[0.13, 0.28]	asm	Large low-wage workforce with automation exposure	M
α_{ret}	0.65	[0.50, 0.78]	der	Wholesale high; retail moderate; combined	M
κ_{ret}	0.12	[0.05, 0.15]	der	OECD 5–15% range	M
λ_{ret}	0.55	[0.40, 0.70]	asm	Large workforce with automation exposure	M
ϕ_{ret}	0.40	[0.28, 0.52]	asm	Moderate enabling exposure	L
η_{ret}	0.30	[0.18, 0.42]	asm	Moderate learning-by-doing	L
μ_{ret}	0.08	[0.04, 0.12]	asm	Moderate skill attrition	L
$\bar{p}_{\text{ret}}$	0.12	[0.05, 0.15]	der	OECD retail 5–15% range	M

13 v0.3 Calibration: Tier 2 Parameters

13.1 Tier 2 Global Rates

Parameter	Value	Unit	Class	Conf.
ρ_A^{T2}	0.28	yr ⁻¹	assumed	low
ρ_P^{T2}	0.20	yr ⁻¹	assumed	low

13.2 Tier 2 Calibration Note: β_s – Absorptive Multiplier

The absorptive multiplier $\beta_s \in [0, 1]$ captures the sector’s fixed structural coupling to the national enabling environment. It replaces the dynamic K_s equation for Tier 2 sectors. A high β_s means the sector responds strongly to improvements in $E(t)$; a low β_s means the sector is relatively insulated from (or unresponsive to) the national enabling environment.

Calibration drivers for β_s :

- **Digital maturity:** sectors with higher baseline digital infrastructure have higher β_s .
- **Enterprise size distribution:** larger-firm sectors have higher β_s (lower adoption barriers per firm).
- **Technology intensity of core operations:** sectors where digital tools are central to daily operations have higher β_s .

- **Regulatory and governance constraints:** regulated sectors may have lower effective β_s even if technically capable.

Values are calibrated using sector archetypes derived from AI Forum NZ, Datacom, and OECD cross-sector evidence, applied to each Tier 2 sector’s characteristics. All Tier 2 β_s values are assumed; no direct NZ observational evidence for this parameter class exists. Sensitivity ranges are wider for Tier 2 than for Tier 1.

13.3 Tier 2 Per-Sector Parameter Tables

S10 – Education and Training

Calibration note. Education is one of the earliest AI adopters in the AI Forum NZ survey (22.4% of respondents) [4], with use cases in personalised learning (University of Otago GPT-4 course recommendations) and administrative automation. However, adoption is uneven across institutions, and barriers include academic integrity concerns, uneven digital infrastructure, and variable staff capability. β_{edu} is set at moderate level reflecting a sector that engages with AI but faces structural institutional constraints.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{edu}}(0)$	0.25	[0.18, 0.35]	der	AI Forum NZ: Education largest respondent group	M
$P_{\text{edu}}(0)$	0.15	[0.10, 0.20]	asm	Derived from $A_{\text{edu}}(0)$	M
β_{edu}	0.35	[0.25, 0.45]	asm	Moderate; institutional constraints on digital adoption	L
κ_{edu}	0.12	[0.08, 0.18]	der	Moderate productivity gains; admin AI primary driver	L
ϕ_{edu}	0.30	[0.20, 0.42]	asm	Moderate labour exposure; augmentation-dominant	L

S11 – Transport and Warehousing

Calibration note. Route optimisation, logistics AI, and predictive maintenance are primary use cases. The sector spans highly digital logistics operators (large firms) and owner-operator trucking (very low digital maturity). AI Forum NZ shows Transport at 2.2% of survey respondents – one of the lowest shares [4]. β_{trn} is set modestly below Education, reflecting the mixed physical-digital nature of the sector.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{trn}}(0)$	0.15	[0.10, 0.22]	asm	Low; 2.2% of AI Forum respondents	M
$P_{\text{trn}}(0)$	0.08	[0.05, 0.12]	asm	Derived from $A_{\text{trn}}(0)$	L
β_{trn}	0.30	[0.20, 0.40]	asm	Moderate; mixed digital/physical operations	L
κ_{trn}	0.15	[0.10, 0.22]	der	Route optimisation and logistics gains	L
ϕ_{trn}	0.35	[0.22, 0.48]	asm	Moderate labour exposure; autonomous vehicles on horizon	L

S12 – Accommodation and Food Services

Calibration note. High labour intensity, small business dominance (26,070 enterprises), and low digital maturity define this sector. AI use is primarily in booking systems, dynamic pricing, and customer service chatbots – all driven by large chain operators. Most firms are too small to have meaningful AI capability. β_{acc} is the lowest of any Tier 2 sector, reflecting limited structural absorption capacity. Labour exposure ϕ_{acc} is set relatively high given the sector’s many routine service tasks.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{acc}}(0)$	0.10	[0.05, 0.16]	asm	Small firms; low digital maturity; limited evidence	L
$P_{\text{acc}}(0)$	0.06	[0.03, 0.10]	asm	Derived from $A_{\text{acc}}(0)$	L
β_{acc}	0.22	[0.15, 0.30]	asm	Low absorptive capacity; fragmented sector	L
κ_{acc}	0.08	[0.05, 0.14]	der	Limited AI applicability; physical hospitality work	L
ϕ_{acc}	0.40	[0.28, 0.52]	asm	High labour exposure; many routine service tasks	L

S13 – Administrative and Support Services

Calibration note. Administrative tasks are among the highest AI-applicability use cases globally (AI Forum NZ: 13.8% of AI deployments are in administration) [4]. The sector provides outsourced admin, facilities management, and business support – activities with high routine content and strong GenAI applicability. AI Forum NZ survey shows 3.9% of respondents from this sector. β_{adm} and ϕ_{adm} are set moderately high relative to other Tier 2 sectors.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{adm}}(0)$	0.25	[0.18, 0.35]	asm	Admin tasks highly automatable; early adopters	M
$P_{\text{adm}}(0)$	0.15	[0.10, 0.20]	asm	Derived from $A_{\text{adm}}(0)$	M
β_{adm}	0.38	[0.28, 0.50]	asm	Moderate capacity; routine task AI applicability high	L
κ_{adm}	0.18	[0.12, 0.25]	der	Higher productivity gains from automation of admin tasks	L
ϕ_{adm}	0.55	[0.40, 0.70]	asm	High labour exposure; many admin tasks automatable	L

S14 – Information Media and Telecommunications

Calibration note. The ICT sector is a digital-native industry with the highest structural AI absorption capacity of any Tier 2 sector. AI Forum NZ shows Information, Media and Telecommunications at 12.1% of survey respondents – second only to Education [4]. The high $\beta_{\text{imt}} = 0.60$ reflects the sector’s fundamental digital orientation. This is a sector where E improvements translate efficiently into adoption.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{imt}}(0)$	0.50	[0.40, 0.62]	asm	Digital-native; naturally high AI adoption	M
$P_{\text{imt}}(0)$	0.30	[0.22, 0.38]	asm	Derived from $A_{\text{imt}}(0)$	M
β_{imt}	0.60	[0.45, 0.72]	asm	High absorptive capacity; digital-native sector	M
κ_{imt}	0.22	[0.15, 0.30]	der	Moderate-high gains from further AI integration	M
ϕ_{imt}	0.35	[0.22, 0.48]	asm	Moderate labour exposure; tech skills augmented	L

S15 – Utilities

Calibration note. Smart grid AI, demand forecasting, predictive asset maintenance, and outage management are primary AI applications. The sector is capital-intensive and highly regulated, which constrains adoption speed. AI Forum NZ shows Electricity, Gas, Water at 2.2% of respondents. β_{utl} is set at moderate level; the regulatory constraint moderates what would otherwise be a technically capable sector. NZ’s electricity sector has special strategic interest in AI for the transition to 100% renewable generation.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{utl}}(0)$	0.20	[0.14, 0.28]	asm	Smart grid AI emerging; regulatory constraints	M
$P_{\text{utl}}(0)$	0.12	[0.08, 0.17]	asm	Derived from $A_{\text{utl}}(0)$	M
β_{utl}	0.32	[0.22, 0.42]	asm	Moderate capacity; regulatory friction applies	L
κ_{utl}	0.18	[0.12, 0.25]	der	Operational efficiency gains in regulated sector	L
ϕ_{utl}	0.25	[0.15, 0.38]	asm	Moderate labour exposure; capital-intensive sector	L

14 v0.3 Calibration: Tier 3 Parameters

14.1 Tier 3 Global Rate

Parameter	Value	Unit	Class	Conf.
ρ_A^{T3}	0.18	yr ⁻¹	assumed	low

14.2 Tier 3 Calibration Note: ψ_s – Productivity Coefficient

The productivity coefficient $\psi_s \in [0, 1]$ translates adoption $A_s(t)$ directly into a normalised productivity effect for Tier 3 sectors. It captures the structural intensity of AI impact on sector output per unit of adoption.

Calibration drivers for ψ_s :

- **Nature of work:** cognitive-intensive sectors have higher ψ_s than predominantly physical sectors.
- **AI applicability:** sectors where AI tools map directly onto core tasks have higher ψ_s .
- **Evidence quality:** where the empirical basis is weaker, ψ_s is set conservatively and carries a wider sensitivity range.

All Tier 3 ψ_s values are assumed. Sensitivity ranges are the widest of any parameter class in the model. These sectors are included for denominator honesty; the productivity contribution of Tier 3 to the whole-economy aggregate should be interpreted with caution.

14.3 Tier 3 Calibration Note: γ_s – Adoption Multiplier

The adoption multiplier $\gamma_s \in [0, 1]$ plays the same structural role in Tier 3 as β_s in Tier 2: it captures the sector’s fixed coupling to the national enabling environment. Tier 3 sectors are expected to have lower γ_s values than the corresponding Tier 2 sectors with similar characteristics, reflecting the residual nature of these sectors and the paucity of evidence for calibrating their responsiveness.

14.4 Tier 3 Per-Sector Parameter Tables

S16 – Mining

Calibration note. NZ mining is a small sector (2,024 enterprises, \$2B GDP). AI applications include geological survey analysis, predictive equipment maintenance, and safety monitoring. International resource extraction AI exists (Rio Tinto, BHP) but NZ-specific evidence is absent. Parameters are set at the lower end of plausible ranges.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\min}(0)$	0.12	[0.06, 0.20]	asm	Small sector; international analogues proxy	L
$\gamma_{\min}$	0.25	[0.15, 0.35]	asm	Moderate coupling; physical extraction sector	L
$\psi_{\min}$	0.12	[0.08, 0.18]	asm	Limited productivity gains; physical tasks dominant	L

S17 – Rental, Hiring and Real Estate

Calibration note. This is the largest Tier 3 sector by GDP (\$40.3B, largely imputed rent from owner-occupied housing). Property technology AI (pricing algorithms, listing optimisation, due diligence automation) applies to the active real estate component. The large GDP weight ($\omega_{\text{rnt}} = 0.1435$) is primarily driven by imputed rent, which does not directly respond to AI adoption. ψ_{rnt} is set low to reflect that most of this sector’s GDP value is in property assets, not AI-responsive activities.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{rnt}}(0)$	0.15	[0.08, 0.22]	asm	Proptech AI emerging; mostly imputed rent sector	L
γ_{rnt}	0.28	[0.18, 0.38]	asm	Moderate coupling; large imputed-rent component	L
ψ_{rnt}	0.10	[0.06, 0.15]	asm	Low; imputed rent component does not respond to AI	L

S18 – Arts and Recreation

Calibration note. Creative AI tools (generative image, audio, text) are directly relevant but face mixed reception in arts communities and potential intellectual property and authenticity concerns. Recreation services (gyms, sports) have limited AI applicability. AI Forum NZ survey shows Other Services (which includes arts) at 4.3% of respondents. ψ_{art} is set slightly higher than other Tier 3 sectors to reflect the genuine creative AI applicability, but with a wide sensitivity range.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{art}}(0)$	0.18	[0.10, 0.28]	asm	Creative AI tools emerging; mixed reception	L
γ_{art}	0.28	[0.18, 0.38]	asm	Moderate coupling; digital content creation pathway	L
ψ_{art}	0.15	[0.10, 0.22]	asm	Moderate; creative tools boost output in suitable sub-sectors	L

S19 – Other Services

Calibration note. A residual sector comprising personal services, repair services, religious and civic organisations, and other non-classified activities. Heterogeneous and diffuse. Limited AI applicability for most sub-sectors. Parameters are set at the most conservative values in the model. This sector is included solely to close the denominator; its model contribution should not be interpreted as an economic claim about AI in Other Services.

Parameter	Value	Sensitivity	Class	Source	Conf.
$A_{\text{oth}}(0)$	0.12	[0.06, 0.18]	asm	Diverse residual sector; very limited evidence	L
γ_{oth}	0.22	[0.12, 0.32]	asm	Lower coupling; heterogeneous personal services	L
ψ_{oth}	0.08	[0.05, 0.12]	asm	Limited productivity gains; physical and personal services	L

15 v0.3 Calibration: Confidence-Adjusted Aggregate

15.1 Motivation

The whole-economy aggregate $\bar{P}(t)$ in Equation (25) weights each sector by its GDP share ω_s but not by the model's confidence in that sector's state representation. Tier 3 sectors contribute to $\bar{P}(t)$ with the same GDP-proportional weight as Tier 1 sectors, despite their substantially lower modelling fidelity and evidential basis.

In v0.3, a confidence-adjusted aggregate is introduced as a complementary output. It is not the primary aggregate; the GDP-weighted $\bar{P}(t)$ remains the canonical whole-economy productivity indicator. The confidence-adjusted variant provides a conservative bound that downweights low-confidence contributions.

15.2 Tier Confidence Weights

The modelling confidence weights by tier are:

Tier	Confidence weight c_s	Rationale
Tier 1	1.00	Full state; sector archetypes; direct evidence base
Tier 2	0.70	Reduced state; fixed absorptive multiplier; limited sector evidence
Tier 3	0.40	Minimal state; ex-post only; included for denominator honesty

15.3 Confidence-Adjusted Aggregate Definition

$$\bar{P}^c(t) = \frac{\sum_{s \in \mathcal{S}_1} \omega_s P_s(t) + \sum_{s \in \mathcal{S}_2} 0.70 \cdot \omega_s P_s(t) + \sum_{s \in \mathcal{S}_3} 0.40 \cdot \omega_s \psi_s A_s(t)}{\sum_{s \in \mathcal{S}_1} \omega_s + \sum_{s \in \mathcal{S}_2} 0.70 \cdot \omega_s + \sum_{s \in \mathcal{S}_3} 0.40 \cdot \omega_s} \quad (26)$$

The denominator normalises $\bar{P}^c(t)$ so that it remains in $[0, 1]$ and is comparable to $\bar{P}(t)$.

Remark 11. *The normalised denominator in Equation (26) equals:*

$$\sum_{s \in \mathcal{S}_1} \omega_s + 0.70 \sum_{s \in \mathcal{S}_2} \omega_s + 0.40 \sum_{s \in \mathcal{S}_3} \omega_s = 0.506 + 0.70 \times 0.217 + 0.40 \times 0.282 = 0.506 + 0.152 + 0.113 = 0.771.$$

The effective GDP coverage in the confidence-adjusted aggregate is 77.1% of the model-consistent total, compared to 100% for the unweighted $\bar{P}(t)$.

Remark 12. *The $\bar{P}^c(t)$ aggregate is expected to be lower than $\bar{P}(t)$ in simulations, because the high-weight Tier 3 sector (S17 Rental/Real Estate, $\omega_{rnt} = 0.1435$) is down-weighted from its GDP-proportional contribution. For policy comparison purposes, the gap between $\bar{P}(t)$ and $\bar{P}^c(t)$ indicates the sensitivity of scenario conclusions to the modelling of low-confidence sectors.*

16 Known Limits of v0.3

This section supplements the structural limits inherited from v0.2 (Section 8) with limits specific to the v0.3 calibration.

1. **No historical fitting.** All parameters are cross-sectionally calibrated using available survey evidence, sector archetypes, and OECD benchmarks. No parameter has been fitted to NZ historical time series. The implied system dynamics have not been validated against observed NZ adoption or productivity data. This is explicitly deferred to v0.4 or a simulation build phase.

2. **Most Tier 1 parameters are assumed, not observed.** Of the 11 structural parameters per Tier 1 sector, only $A_s(0)$ has a partial observed or derived basis for some sectors. The remaining 10 parameters are assumed, though derived from sector archetypes informed by OECD and NZ research. All assumed parameters carry wide sensitivity ranges.
3. **All Tier 2 β_s and all Tier 3 γ_s, ψ_s are assumed.** No NZ-specific observational or derived evidence exists for the Tier 2 absorptive multipliers, Tier 3 adoption multipliers, or Tier 3 productivity coefficients. These parameters are the highest-priority targets for evidence improvement in future versions.
4. **GDP share weights use model-sector definitions, not pure ANZSIC Level 1.** Three adjustments were made to align the model’s 19 sectors with non-overlapping GDP coverage: splitting S03/S13, deriving S05 Technology, and splitting S18/S19. These adjustments introduce derived uncertainty into the ω_s values for those sectors. The mapping methodology is documented in Section 10.
5. **Confidence tier weights (c_s) are assumed.** The values 1.00, 0.70, and 0.40 for Tiers 1, 2, and 3 are transparent assumptions with no formal evidential basis. They serve a directional purpose: to show what happens to the aggregate when low-confidence contributions are reduced. Different reasonable choices of c_s would change $\bar{P}^c(t)$ quantitatively but not qualitatively.
6. **Parameter table is partially filled.** Of the 348 cells in `data/sector-parameter-table.csv`, approximately 120 are filled as of v0.2. v0.3 fills additional Tier 2 and Tier 3 model-calibration cells. The data requirement for a fully validated simulation (especially Tier 1 C2–C7 fields, D1–D7 fields, Tier 2 B2–B3 and C1–C2 fields) remains a gap.

17 References

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